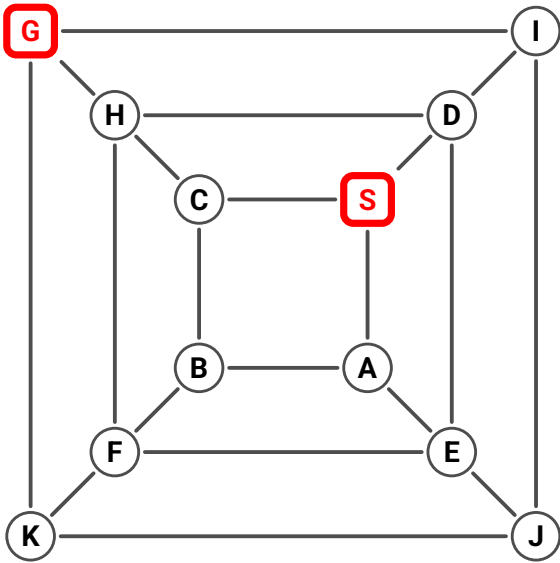


Problem 6

Depth First Search

Prepared by S. Baskaran

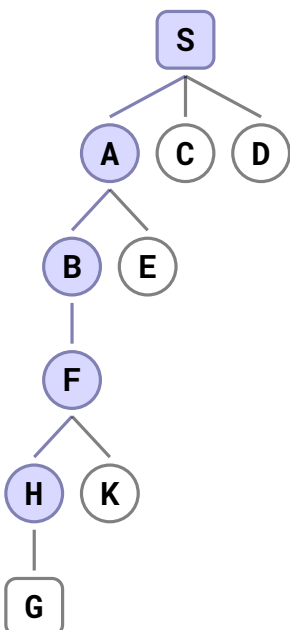
State Space



MoveGen returns nodes in ALPHABETICAL order.

- S -> A, C, D
- A -> B, E, S
- B -> A, C, F
- C -> B, H, S
- D -> E, H, I, S
- E -> A, D, F, J
- F -> B, E, H, K
- H -> C, D, F, G
- I -> D, G, J
- J -> E, I, K
- K -> F, G, J
- G -> H, I, K

Depth First Search – Search Tree



Depth First Search – Solution

OPEN and CLOSED carry pairs: (NODE,PARENT)

OPEN (S,null):[]
CLOSED []

1.
NODE S
close (S,null)
moveGen A:C:D:[]
newNodes A:C:D:[]
newPairs (A,S):(C,S):(D,S):[]

OPEN (A,S):(C,S):(D,S):[]
CLOSED (S,null):[]

2.
NODE A
close (A,S)
moveGen B:E:S:[]
newNodes B:E:[]
newPairs (B,A):(E,A):[]

OPEN (B,A):(E,A):(C,S):(D,S):[]
CLOSED (A,S):(S,null):[]

3.
NODE B
close (B,A)
moveGen A:C:F:[]
newNodes F:[]
newPairs (F,B):[]

OPEN (F,B):(E,A):(C,S):(D,S):[]
CLOSED (B,A):(A,S):(S,null):[]

4.
NODE F
close (F,B)
moveGen B:E:H:K:[]
newNodes H:K:[]
newPairs (H,F):(K,F):[]

OPEN (H,F):(K,F):(E,A):(C,S):(D,S):[]
CLOSED (F,B):(B,A):(A,S):(S,null):[]

5.
NODE H
close (H,F)
moveGen C:D:F:G:[]
newNodes G:[]
newPairs (G,H):[]

OPEN (G,H):(K,F):(E,A):(C,S):(D,S):[]
CLOSED (H,F):(F,B):(B,A):(A,S):(S,null):[]

6.
NODE G
GOAL G

OPEN (G,H):(K,F):(E,A):(C,S):(D,S):[]
CLOSED (H,F):(F,B):(B,A):(A,S):(S,null):[]

PATH S:A:B:F:H:G:[]